FRAUD ANALYSIS REPORT

Executive Takeaway

Prioritize review of **mobile credit-card transactions**, especially categories **C and S**. Within this segment, **Mastercard (16.66%)** and **Visa (13.77%)** show the highest displayed card-network fraud rates. Overall, categories **W (11.69%)** and **C (5.90%)** stand out, while **Discover (7.73%)** has the highest displayed overall network fraud rate. **Proton email addresses (40.79%)** also show a high observed fraud rate.

Key Metrics

590,540	20,663	3.50%	$79.74M	$3.08M
Total Transactions	Fraud Transactions	Fraud Rate	Total Amount	Fraud Amount

Key Insights

1. Overall card categories: **W (11.69%)** and **C (5.90%)** have the highest displayed fraud rates.

2. Overall card networks: **Discover (7.73%)** leads, followed by **Visa (3.47%)**, **Mastercard (3.43%)**, and **American Express (2.87%)**.

3. Mobile + credit: **13.91% fraud rate**, with **3,312 fraud transactions** out of **23,802 total transactions**. Category **C (20.70%)**, **S (19.32%)**, and **R (8.71%)** stand out.

4. Mobile + credit card networks: **Mastercard (16.66%)** and **Visa (13.77%)** show the highest displayed fraud rates; Discover is **9.76%** and American Express is **5.44%**.

5. **Visa and Mastercard** account for the largest displayed total transaction amounts overall. Review transaction value alongside fraud amount, fraud count, and fraud rate.

6. **Proton email addresses (40.79%)** have the highest displayed overall email-domain fraud rate. Confirm transaction volume before making policy decisions.

Recommendations

1. Prioritize review of **mobile credit-card transactions**, starting with categories **C and S**, then category R.

2. Review elevated-risk **Mastercard and Visa** activity, especially high-value, repeated, or unusual transactions.

3. Investigate overall categories **W and C** and compare fraud rates with transaction volumes.

4. Use **Proton email domain** as one risk signal alongside device, amount, and transaction history—not as a standalone blocking rule.

5. Validate sample sizes and fraud-rate calculations, and monitor false positives before applying new controls.

Decision

Approve a focused review of mobile credit-card transactions, prioritizing categories C and S and elevated-risk Mastercard/Visa activity, while monitoring overall categories W and C and investigating Proton-domain patterns.